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Key Points:

- Flow and transport through an ensemble of networks with varying densities are simulated
- Novel measurements of flow channeling degree are introduced
- As the network connectivity increases, the degree of network-scale flow channeling decreases

Correspondence to:

J. D. Hyman,
jhyman@lanl.gov

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Flow Channeling in Fracture Networks: Characterizing the Effect of Density on Preferential Flow Path Formation

Jeffrey D. Hyman¹ 

¹Computational Earth Science Group (EES-16), Earth and Environmental Sciences Division, Los Alamos National Laboratory, Los Alamos, NM, USA

Abstract Flow channelization is a commonly observed phenomenon in fractured subsurface media where the flow of fluids is restricted primarily to highly transmissive fracture networks surrounded by a low-permeability rock matrix. The multiscale structural heterogeneity of these networks results in multiscale flow channelization where preferential flow paths form at length scales ranging from the entire system down to the subfracture size. We present an analysis of how one of the largest scales in fractured media, the network density, influences the degree of flow channeling that occurs using an ensemble of semigenetic three-dimensional discrete fracture network (DFN) simulations. We construct 10 DFNs, whose fracture lengths follow a power law distribution, at four densities for a total of 40 networks. We characterize their structure in terms of the network topology and geometry. Eulerian and Lagrangian observations of the steady-state flow fields obtained within the networks are used to quantify the degree of flow channelization at the network scale. We introduce a measure for the importance-ranking/hierarchy of different flow paths in the network using graph-based analysis of Lagrangian transport by which the degree of flow channeling between networks is compared. These flow observations are then linked to the structural properties of the networks. In general, network-scale flow channeling decreases as the network density increases. However, at low densities, there is more uniform flow within the entire connected network than in high-density networks. We also demonstrate how standard transport observables can be used to infer the degree of flow channelization occurring within a fracture network.

1. Introduction

Nonuniform fluid velocity fields have been observed in laboratory (Datta et al., 2013; Gentier et al., 1989; Hakami & Larsson, 1996; Jiménez-Martínez et al., 2017, 2020; Morales et al., 2017; Su et al., 2001) and field experiments (Abelin et al., 1985, 1991; Frost et al., 1992; Guihéneuf et al., 2014; Le Borgne et al., 2006; Rasmuson & Neretnieks, 1986; Winograd & Pearson, 1976) as well as numerical simulations (Cirpka et al., 2015; Cvetkovic et al., 2004; de Dreuzy et al., 2012; Frampton et al., 2019; Grindrod & Impey, 1993; Hyman & Winter, 2013; Hyman et al., 2012; Kang et al., 2014; Le Goc et al., 2010; Tompson & Gelhar, 1990) of flows through heterogeneous and fractured porous media at scales ranging from microns to kilometers. The medium heterogeneity offers spatially variable resistance to flow and a diverse fluid velocity field that contains *flow channels*, connected regions of high fluid velocity also known as preferential flow paths, form therein. While it is commonly assumed that the regions where flow channels form are those that provide the least resistance to flow, recent studies have shown that a more nuanced approach is warranted. In contrast to this conventional wisdom, the regions where channels form are nonunique. They can depend dynamically the direction of the driving force (Grindrod & Impey, 1993; Moreno & Tsang, 1994; Neuman, 2005; Tsang & Tsang, 1989), multiphase fluid properties (Birkholzer & Tsang, 1997; Datta et al., 2013; Flury et al., 1994; de Gennes, 1983; Hyman et al., 2020; Jiménez-Martínez et al., 2016), and connectivity of the permeability field (Edery et al., 2014; Knudby & Carrera, 2005; Renard & Allard, 2013; Rizzo & de Barros, 2017; Tyukhova & Willmann, 2016; Zinn & Harvey, 2003). Classical models, such as the advection-dispersion equation, inadequately represent/predict solute transport through systems with a high degree of flow channelization because these features result in transport behavior that exhibits early arrival or power law scaling in the tail of the travel time distributions; that is, it is non-Fickian (Berkowitz & Scher, 1997; Cushman & O'Malley, 2015; Edery et al., 2014; Painter et al., 2002). In turn, our predictive modeling capabilities are severely limited in many subsurface engineering endeavors including aquifer

management, hydrocarbon extraction, and the long-term storage of spent nuclear fuel (Follin et al., 2014; Hyman et al., 2016; Jenkins et al., 2015; Kueper & McWhorter, 1991; Middleton et al., 2017; National Research Council, 1996; Neuman, 2005; Selroos et al., 2002; VanderKwaak & Sudicky, 1996).

These preferential pathways are particularly pronounced in low-permeability fractured rock, for example, granites and shales, where the flow of fluid and the associated transport of dissolved chemicals are primarily confined to interconnected networks of fractures surrounded by a low-permeability rock matrix (cf. Tsang & Neretnieks, 1998, for an overview of flow channeling through heterogeneous fractured rocks in laboratory and field experiments). Within these subsurface systems, individual fractures connect to form complicated networks where relevant length scales can span several orders of magnitude (Bonnet et al., 2001). This multiscale structural heterogeneity leads to multiscale flow channelization at length scales ranging from the entire system down to the subfracture size. At the largest scale, network density and connectivity alter the distribution of flow within the network, but this connection has only been qualitatively identified at this time (Huseby et al., 2001; Hyman & Jiménez-Martínez, 2018; Hyman et al., 2019a; Sherman et al., 2020; Sweeney & Hyman, 2020). While it is generally accepted that a lower degree of flow channeling ought to occur in higher-density networks, how to quantify the degree of flow channeling and link those measurements to geological features remains unclear. For example, Sherman et al. (2020) observed that network density affected flow channeling and then studied the ability of upscaled transport models to reproduce high-fidelity transport simulations using DFN models at these various densities. However, they did not systematically study the link between network structure attributes and flow channel formation nor develop new methods to compare flow channelization between networks. At the scale of individual fractures, their orientations (Rubin & Buddemeier, 1996) and apertures (Bogdanov et al., 2007; de Dreuzy et al., 2002; Frampton & Cvetkovic, 2010; Hyman et al., 2016; Joyce et al., 2014; Wellman et al., 2009) can also cause variations of the flow field. At the even smaller scale within a single fracture, aperture variability/roughness is well known to lead to flow channelization (Boutt et al., 2006; Cardenas et al., 2007; Detwiler et al., 2000; de Dreuzy et al., 2012; Frampton et al., 2019; Kang et al., 2016; Makedonska et al., 2016; Méheust & Schmittbuhl, 2000, 2001). In summation, the integration of these multiple scales renders linking structural attributes to the formation of flow channels a particularly challenging task.

Although flow channelization is a straight forward phenomenon to describe qualitatively, there is a lack of quantitative measurements to evaluate the degree of flow channeling being observed (Le Borgne et al., 2006, 2007; Su et al., 2001). This need for adequate and appropriate measurements of flow channelization has made it extremely difficult to link its occurrence with geological properties. The primary method used to identify preferential flows and the regions where they occur is critical thresholding (Ambegaokar et al., 1971; Kirkpatrick, 1971; Otsu, 1979), where parts in the domain are excluded if the value there is lower than a given threshold, and that threshold value is iteratively lowered until a subdomain that connects boundaries; that is, a percolating path, is identified (Berkowitz & Balberg, 1993). Although this methodology has proven fruitful in its applications to heterogeneous porous media (cf. Bianchi et al., 2011; Edery et al., 2014; Hasan et al., 2019) for recent examples, the extension to fracture networks is not straight forward because one must consider constraints imposed by the network structure in addition to the hydrological resistance. Recent work has focused on integrating connectivity with medium properties and appears promising (Hyman et al., 2017, 2018; Rizzo & de Barros, 2017; Tyukhova & Willmann, 2016). In conjunction with the recent advances in numerical simulations of flow and transport in fractured media, these methods will allow for the impact of geophysical properties on flow channelization to be identified in more quantitative terms (Berre et al., 2019). To date, however, most of the numerical studies concerning flow channeling in fractures have focused on either single fractures (Boutt et al., 2006; Cardenas et al., 2007; Detwiler et al., 2000; Kang et al., 2016) or two-dimensional networks (Edery et al., 2016; Kang et al., 2015, 2017; Le Goc et al., 2010). Nonetheless, an improved understanding of this connection between geophysical properties and flow channeling is essential for model selection of flow and transport in fractured rock ranging from the use of upscaled continuum models to the application of theoretical non-Fickian transport models, such as the continuous-time random walk (CTRW) or fractional-order transport models (fADE) (Benson et al., 2000; Berkowitz et al., 2006; Bolster et al., 2014; Dentz & Berkowitz, 2003; Dentz et al., 2016; Kang et al., 2011; Le Borgne et al., 2008; Neuman & Tartakovsky, 2009).

In this study, we perform a set of three-dimensional discrete fracture network (DFN) simulations designed to isolate the impact of network density, one of the largest scales present in fracture networks, on the formation of preferential flow paths. The high fidelity of the simulations allows for the detailed examination of the flow field within each fracture network, thereby providing a basis to develop quantitative measurements of the degree of flow channeling, which is a principal contribution of this work. We use a DFN model rather than a continuum representation because fractures are explicitly represented in a DFN, which allows us to link their attributes directly with flow and transport phenomena (Hadgu et al., 2017; Painter & Cvetkovic, 2005; Sweeney et al., 2020). The networks are semigeneric, meaning that they do not represent a particular field site, but their properties are motivated by field observations. The use of semigeneric DFNs allows for the development and exposition of the proposed framework to measure and compare flow channeling without site-specific considerations, which thereby increases their generality and applicability. We construct 10 DFNs, whose the fracture lengths follow a power law distribution, at four densities for a total of forty networks. The structure of each DFN is characterized in terms of its network topology and fracture geometry. Steady-state flow simulations are performed to obtain the fluid velocity field within each fracture, and particle tracking is used to simulate passive transport through the networks. To address the need for more quantitative measurements of flow channeling, we develop a suite of Eulerian and Lagrangian flow properties that quantify the degree of flow channeling in each DFN. Specifically, we introduce a novel measure for the importance ranking/hierarchy of different flow paths in the network using graph-based analysis of Lagrangian transport. In summation, these simulations and novel methods to measure flow channeling allow us to identify direct links between flow channeling and geophysical attributes. We then described the effects of these flow channels on particle transport properties, including tortuosity and particle breakthrough times.

2. Numerical Simulations

In this section, we describe our modeling methodology to simulate and analyze flow and transport through subsurface fracture networks.

2.1. Discrete Fracture Networks

We use the high-fidelity three-dimensional discrete fracture network modeling suite *dfnWorks* (Hyman, Karra, et al., 2015) to generate each DFN, solve the steady-state flow equations, and simulate transport therein using particle tracking. *dfnWorks* combines the feature rejection algorithm for meshing (*fram*) (Hyman et al., 2014), the *LaGriT* meshing toolbox (*LaGriT*, 2013), the parallelized subsurface flow and reactive transport code *pflotran* (Lichtner et al., 2015), and an extension of the walkabout particle tracking method (Hyman et al., 2019; Makedonska et al., 2015; Painter et al., 2012). *fram* is used to generate three-dimensional fracture networks. *LaGriT* is used to create a computational mesh representation of the DFN in parallel. *pflotran* is used to numerically integrate the governing flow equations. *walkabout* is used to determine pathlines through the DFN and simulate solute transport that can consider matrix diffusion as well as purely advective transport. Details of the suite, its abilities, applications, and references for detailed implementation are provided in Hyman, Karra, et al., (2015).

2.1.1. Network Generation

We consider an ensemble of networks with various densities. The generation parameters of the networks are all the same, except the number of fractures included in the domain. We consider a single fracture family whose centers are uniformly distributed throughout the domain based on a stochastic process. This style DFN generation is referred to as a Poissonian generation opposed to a kinematic generation where fractures nucleate and grow (Davy et al., 2010, 2013; Lavoine et al., 2020; Thomas et al., 2020). The fracture orientations are sampled from a Fisher distribution with intensity parameter $\kappa \approx 0$ and mean normal vector of (0,0,1) using the method provided by Wood (1994). This method results in a uniform covering of the unit sphere, which mimics disordered fractures networks observed in the field (Klint et al., 2004). Fractures are represented as planar discs whose radii r follow a power law distribution with decay exponent α , which is a commonly observed property of fracture networks in the field (Bonnet et al., 2001). For computational purposes, we use a truncated power law distribution with upper and lower cutoffs (r_u ; r_0) and exponent α :

$$p_r(r, r_0, r_u) = \frac{\alpha}{r_0} \frac{(r/r_0)^{-1-\alpha}}{1 - (r_u/r_0)^{-\alpha}}. \quad (1)$$

We consider an exponent of 1.8, a lower cutoff of 1 m and upper cutoff of 10 m. Each network is generated in a cubic domain with sides of length $L = 50$ m. We select these values to ensure multiple fractures must connect between the inflow and outflow boundaries. During the generation of the network, we expand the domain slightly (5 m) in each direction to avoid low-density issues near domain boundaries, which can occur due to sampling bias. We nondimensionalize all length scales by the minimum fracture size r_0 .

The feature rejection algorithm for meshing (FRAM) is used to create an unstructured conforming Delaunay triangulation of each DFN (Hyman et al., 2014). We use a spatially variable mesh resolution where the mesh is refined close to the intersections and gradually coarsened away from them using the algorithm presented in Ushijima-Mwesigwa et al. (2019); the mesh resolution is a piecewise linear function of distance from intersections on each fracture plane. Gradients in the flow field are highest close to the fracture intersections, which is why we refine the mesh in these regions. We set the minimum mesh resolution to $0.01 r_0$.

Variations hydraulic properties between fractures are included into the networks by positively correlating hydraulic aperture to the fracture radii using a power law relationship $b = 5.0 \times 10^{-4} \sqrt{r}$. The parameters of the relationship are based on field data (Svensk Kärnbränslehantering AB, 2010). This correlation between fracture size and aperture is a common assumption in DFN models (Bogdanov et al., 2007; Frampton & Cvetkovic, 2010; Hyman et al., 2016; Joyce et al., 2014; Wellman et al., 2009), which has been shown to adjust flow velocity within principal channels but not significantly rearrange the flow field structure (de Dreuzy et al., 2002; Hyman et al., 2016). Under this model setup, the hydraulic aperture is constant within each fracture, but it varies between fractures with larger fractures having wider hydraulic apertures than smaller fractures. Although it is well documented that variations in aperture roughness within a fracture lead to flow channeling within an individual fracture (Boutt et al., 2006; Cardenas et al., 2007; Detwiler et al., 2000; Kang et al., 2016) and it is possible to include in-fracture aperture variations into three-dimensional DFN simulations (cf. de Dreuzy et al., 2012; Frampton et al., 2019; Makedonska et al., 2016), we do not do so here. The principal focus of this study is characterizing how variations in the network structure at the macroscale induce flow channeling. In subsequent studies, we plan to investigate the relative influence of the other relevant length scales on flow channeling, including how microscale variations in aperture roughness can manifest at the network scale.

The density of the fracture networks is measured using the percolation parameter p defined by de Dreuzy et al. (2000). De Dreuzy et al. (2012) provided a generalization of the theory presented in de Dreuzy et al. (2000) to include upper and lower cutoffs in the truncated power law distribution based on the truncated third moment of the fracture size distribution

$$p = \frac{N}{L^2} \int_{r_0}^{r_u} dr \min(r, \alpha L) p_r(r). \quad (2)$$

We define the critical percolation density value (denoted p_c) to be the minimum number of fractures required, so there is almost surely a connected cluster of fractures than spans the whole domain, that is, the value of N such that p in (2) is equal to unity (Berkowitz & Balberg, 1993; Bour & Davy, 1997, 1998; Sahimi, 1994). For the domain size and truncated power law distribution parameters considered here, $p_c \approx 750$ fractures. The advantage of this definition of density is that all networks are placed into a relative context using a dimensionless form of the percolation parameter $p' = p/p_c$, where p is the number of fractures in a particular network during generation. We consider values of $p' = 2, 3, 5$, and 10 . Ten networks are generated at each density. We consider both the entire network and the subnetwork connecting inflow and outflow boundaries, where isolated clusters of fractures are removed because they do not participate in flow or transport. Detecting clusters that span the domain and isolated clusters is performed using a graph-based method within dfnWorks (Hyman et al., 2017). One network from each set is shown in Figure 1. Fractures are colored by their radius to demonstrate the wide range of length scales in the simulations.

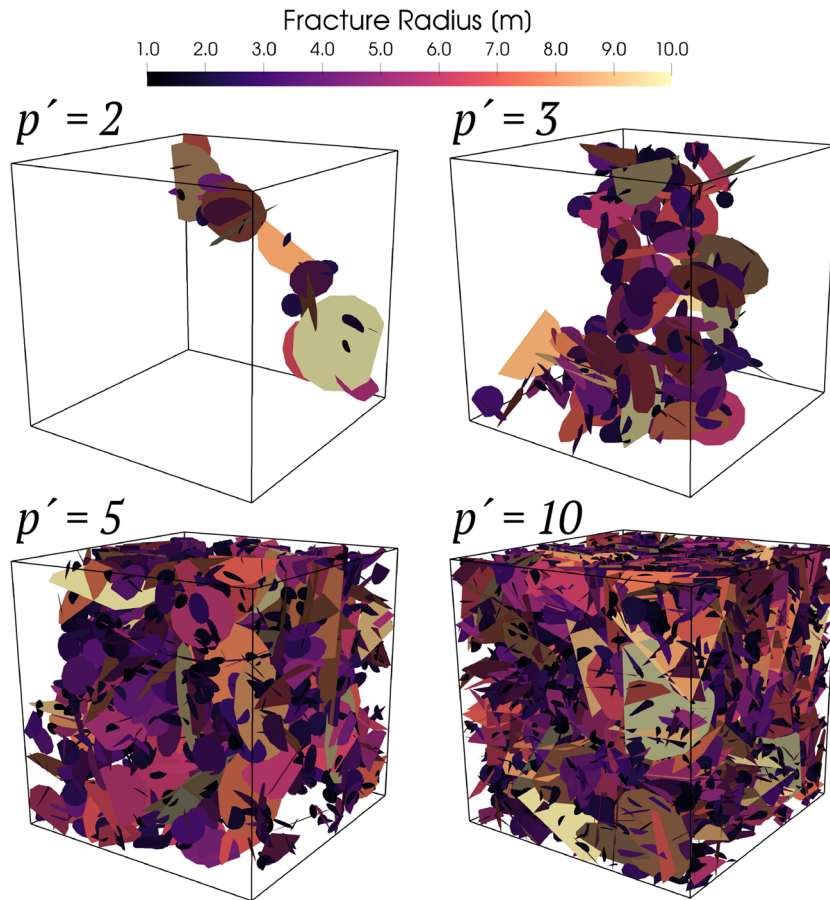


Figure 1. Network realizations from each density. Each network is composed of disc-shaped fractures that connect between the inflow (left) and outflow (right) boundaries. Fractures are colored by their radius.

2.2. Flow and Transport Simulations

Flows through low-permeability fractured rock are typically laminar, and the Stokes equations are an appropriate model to resolve fluid dynamics (Bear, 1988). Due to the large contrast between fracture length and hydraulic aperture, typically several orders of magnitude, the Stokes equation to be reduced to the Reynolds equation (Zimmerman & Bodvarsson, 1996)

$$\nabla \cdot (b^3 \nabla P) = 0. \quad (3)$$

We apply a pressure difference of 1kPa across the domain aligned with the x axis, and all other boundaries are nonflow. We discretize (3) using an unstructured two-point flux finite volume scheme implemented in pflotran to numerically integrate (3) with the prescribed boundary conditions and determine the distribution of pressure and volumetric flow rates on the mesh representation of the DFN generated using FRAM. The matrix surrounding the fractures is assumed to be impervious, and there is no interaction between flow within the fractures and the solid matrix, that is, matrix diffusion is not considered in these simulations. Using the obtained pressure and volumetric flow rates, we reconstruct the Eulerian velocity field $\mathbf{u}(\mathbf{x})$ within the DFN using the method provided in Makedonska et al. (2015) and Painter et al. (2012). The high-resolution mesh allows us to resolve a spatially variable flow field within each fracture plane.

Transport through the network is simulated using a plume of passive, indivisible, and purely advective particles that trace pathlines through the velocity field $\mathbf{u}(\mathbf{x})$. Particles are injected over a plane Ω_a at the inlet plane perpendicular to the mean flow direction. The initial particle positions are denoted $\mathbf{a}$. The distribution of particles along the inlet plane is assigned using flux weighting, which means the number of particles at a

position is proportional to the percentage of the total inflow flux occurring at that position (Hyman, Painter, et al., 2015; Kang et al., 2017; Kreft & Zuber, 1978). One hundred thousand particles are used in each transport simulation. A preliminary study showed that observables were not affected by increasing the number of particles beyond this value.

The trajectory $\mathbf{x}(t; \mathbf{a})$ of a particle starting at $\mathbf{a}$ at time $t = 0$ is given by the advection equation

$$\frac{d\mathbf{x}(t; \mathbf{a})}{dt} = \mathbf{v}_t(t; \mathbf{a}), \quad \mathbf{x}(0; \mathbf{a}) = \mathbf{a}, \quad (4)$$

where the Lagrangian velocity $\mathbf{v}_t(t; \mathbf{a})$ is given in terms of the Eulerian velocity $\mathbf{u}(\mathbf{x})$ as $\mathbf{v}_t(t; \mathbf{a}) = \mathbf{u}[\mathbf{x}(t; \mathbf{a})]$. The dynamics occurring within intersections are a subgrid scale process that is not explicitly resolved and assumed to be diffusion dominated. Under these conditions, the appropriate representation is a complete mixing model where that the probability of exiting into a fracture is weighted by the outgoing flux (Berkowitz et al., 1994; Kang et al., 2015; Park et al., 2001, 2003; Sherman et al., 2019; Stockman et al., 1997).

2.3. Simulation Observables

We compare differences between the network structures and the steady-state flow fields within them using topological, geometric, and physical observables. Topological observables are attributes of the networks related to connectivity, for example, the average number of intersections on a fracture. Geometric observables are those features that can be directly measured without performing a flow or transport simulations, for example, the total surface area of the network or average distance between intersections on a fracture plane. Physical observables are attributes that require simulation of a physical process, for example, distribution of particle travel times through the network.

2.3.1. Topological Observables

We investigate the network connectivity using a graph representation of the networks, where nodes in the graph correspond to individual fractures, and there is an edge between nodes if the corresponding fractures intersect (Huseby et al., 1997; Hyman et al., 2018). We augment the graph to include source and target nodes, corresponding to the inflow and outflow boundaries, to provide a point of reference in the graph with respect to inflow and outflow boundaries. For every fracture that intersects the inflow boundary, an edge is added between the node in the graph corresponding to that fracture and the node representing the inflow boundary, likewise for the outflow boundary. Similar graph-theoretical approaches have been used for a variety of studies concerning fractured media including topological characterization of networks (Andresen et al., 2013; Hope et al., 2015; Huseby et al., 1997; Hyman et al., 2019b), backbone identification (Aldrich et al., 2017; Valera et al., 2018), and uncertainty quantification (Berrone et al., 2020; O'Malley et al., 2018; Osthus et al., 2020). The utility of this graph-based approach is that the topological properties of the networks can be queried and characterized in a formal mathematical framework while retaining physical interpretation.

We consider five topological attributes of the networks. The first two are the mean ($\mu(d)$) and variance ($\sigma(d)$) of the *degree distribution*. In the adopted graph representation, the degree of a node is the number of intersections on the corresponding fracture. The third is the *assortativity coefficient* ($\mathcal{P}$) measured using the Pearson correlation coefficient (Newman, 2002, 2003). Value of $\mathcal{P}$ ranges between -1 and 1 . Values greater than 0 indicate a correlation between nodes of a similar degree, and values less than 0 indicate a correlation between nodes of different degrees. The fourth is node connectivity (n_c). The *node connectivity* of a graph is the fewest number of nodes that need to be removed to disconnect the source from the target. This value provides a measure of structurally enforced channelization in the graph. Lower values indicate that all information passing from source to target must pass through the same nodes. Finally, we measure the *Shannon Entropy* of the networks (H) that corresponds to the ease of information (here flow and transport) to pass through a graph/DFN (Dehmer & Mowshowitz, 2011; Jiménez-Martínez & Negre, 2017; Tutzauer, 2007; Wong & You, 1985). We use the definition where $H = -\sum p(c) \log_2(p(c)^{-1})$ and $p(c)$ is the distribution of degree centrality in the graph.

2.3.2. Geometric Observables

We measure network surface area in terms of the fracture intensity $[r_0^{-1}]$, which is the sum of surface area of each fracture (S_f) divided by the volume of the domain,

$$P_{32} = \frac{1}{V} \sum_f S_f, \quad (5)$$

and is often referred to as P_{32} (Dershowitz & Herda, 1992). We consider the network fracture intensity before and after isolated fractures have been removed, which we denote P_{32} and $\widehat{P}_{32}$ respectively. Next, we measure the distribution of fracture radii that form the connected network and compare it to the prescribed distribution (1). Finally, we measure the distance between intersections on the fracture planes, which is a first-order approximation of the distance that a particle can travel on a fracture before transitioning to another fracture.

2.3.3. Physical Observables

We consider physical observables of the flow field within each network from both Eulerian and Lagrangian perspectives. From the Eulerian point of view, we use the flow channeling density indicator

$$d_Q = \frac{1}{V} \cdot \frac{(\sum_f S_f \cdot Q_f)^2}{(\sum_f S_f \cdot Q_f^2)}, \quad (6)$$

introduced by Maillot et al. (2016) to measure the total surface area where there is significant flow. d_Q weighs the surface area of each fracture by one half the absolute value of total flow exchanged by a fracture with its neighbors (Q_f) in the participation ratio developed in solid state physics (Bell et al., 1970; Edwards & Thouless, 1972). Similar definitions of the participation ratio have previously been applied in the geosciences (Andrade et al., 1999; Davy et al., 1995, 2010). Comparing (5) with (6) suggests that d_Q is a measure of *active* or *flowing* P_{32} . Thus, the percentage of the total surface area where there is significant flow can be quantified using the ratio d_Q/P_{32} . We consider both the ratio of d_Q/P_{32} and $d_Q/\widehat{P}_{32}$.

From the Lagrangian point of view, we consider attributes of the particle plume passing through each network. Some are used to measure flow channeling, while others shall be used to examine the effect of flow channeling on transport behavior. The length of each particle pathline (ℓ) can be used to parameterize the spatial and temporal coordinates of the particle starting at $\mathbf{a}$ to identify the pathline distance where the particle exits the domain

$$\lambda(x_L; \mathbf{a}) = \min\{\ell | x_1(\ell; \mathbf{a}) \geq x_L\}. \quad (7)$$

In turn, the value of $\lambda(x_L; \mathbf{a})$ is used to identify the first arrival time $\tau(x_L, \mathbf{a})$ of a particle at the outlet plane located at x_L

$$\tau(x_L; \mathbf{a}) = \int_0^{\lambda(x_L; \mathbf{a})} \frac{d\ell}{\|\mathbf{v}_t(\ell)\|}. \quad (8)$$

The mass represented by each particle $m(\mathbf{a})$, the total mass of the plume M , and $\tau(x_L, \mathbf{a})$ are combined to compute the relative solute mass flux $\psi(t, x_L)$ breakthrough at a time t ,

$$\psi(t, x_L) = \frac{1}{M} \int_{\Omega_a} d\mathbf{a} m(\mathbf{a}) \delta[\tau(x_L, \mathbf{a}) - t], \quad (9)$$

where $\delta(t)$ is the Dirac delta function. Under the inflow conditions considered here, all particles are assigned the same mass, but their initial positions are distributed according to the inflow flux. We also measure the tortuosity of each particle defined as the length of the pathline compared to the linear distance between inflow and outflow boundaries,

$$\chi(x_L; \mathbf{a}) = \lambda(x_L; \mathbf{a})/x_L. \quad (10)$$

Along each trajectory, we keep track of the fractures that a particle passes through and use this information to construct a flow topology graph. The flow topology graph can be used to identify the fractures and subnetworks where most particles pass through (Aldrich et al., 2017). We begin with the graph representation of

the DFN described in section 2.3.1 and embed particle information as node and edge weights. The node weight is the percentage of particles that pass through the corresponding fracture (π_f), and the weight of an edge is the percentage of particles that pass through the corresponding intersection between fracture i and j ($\pi_{i,j}$).

The flow topology graph allows us to define a particle-based version of the flow channeling indicator, (6). Using the number of particles that pass through each fracture, we compute the particle-based fracture participation ratio:

$$\pi = \frac{1}{N} \frac{(\sum_f \pi_f)^2}{\sum_f \pi_f^2}, \quad (11)$$

where N can be either the total number of fractures in the entire network, prior to isolated clusters being removed (π) or only the connected network ($\hat{\pi}$). We do not weigh π_f by S_f nor normalize the ratio by volume of the domain as this measure is designed to be indicator-based rather than geometrically based.

Next, we introduce a path-based metric using the flow topology graph. First, we define an additional edge weight that is the reciprocal of the number of particles passing between fractures, $1/\pi_{i,j}$. Then we identify the k shortest paths from the source to the target using this weight. The k shortest path is a generalization of the shortest path; they are the k total paths (possibly overlapping) in order of nondecreasing length starting from the shortest path. We denote the k th path in this list as p^k . These k shortest paths correspond to the set of fractures where the largest amount of flow occurs because all particles are passive tracers in the flow and represent the same amount of mass. We consider only simple paths (loopless) from the source to the target.

Once each path has been identified, we define an integrated flow rate along p^k to be the reciprocal of this length of the path multiplied by the number of fractures in that path (n),

$$\pi^k = n \left(\sum_{\{i,j\} \in p^k} \frac{1}{\pi_{i,j}} \right)^{-1}. \quad (12)$$

Notice that the value of π^k is the harmonic average number of particles passing through p^k . The use of the harmonic mean is motivated by conceptualizing the flow topology graph as analogous to flows on a resistor network or obtaining an average hydraulic conductivity value for flow perpendicular to layers. The ordered values of π^k provide an importance ranking/hierarchy of different flow paths in the network. If the values of π^k are relatively constant for increasing values of k , then it indicates that the flow channels all carry a similar amount of flow. On the other hand, if the values of π^k decay rapidly with increasing values of k , then it indicates that there are a few principal channels where most of the mass is being transported.

3. Results

In this section, we report the measurements presented in the previous section obtained in our ensemble of networks. We begin with the network structure and the topological and geometric attributes, then report measurements of flow channeling, and conclude with the impact of the degree of flow channeling on upscaled transport observables.

3.1. Network Characterization

The number of fractures required at each density (N), the final number of fractures in the connected network ($\hat{N}$), and the percentage of the total number of fractures produced during generation (%) are reported in Table 1. The lower densities have fewer fractures than higher densities because all other network parameters are unchanged. These smaller values decrease the probability of fractures to connect and form a network between inflow and outflow boundaries. In the $p' = 2$ and $p' = 3$ network, less than 10% of the total number of fractures are in a cluster connecting boundaries. As the density increases, the difference between the requested and final number of fractures, that is, the number of fractures removed from the domain relative to the required value, decreases, indicating that the networks are better connected at higher density.

Table 1
Network Characterization—Density and Topology

p'	Density			Topology				
	N	$\hat{N}$	%	$\mu(d)$	$\sigma(d)$	$\mathcal{P}$	n_c	H
2	1500	60.20	4%	2.07	2.64	-0.37	1.00	5.43
3	2250	220.10	9%	2.20	3.49	-0.25	1.2	7.26
5	3750	1339.90	35%	2.32	4.95	-0.18	6.9	9.92
10	7500	4069.30	54%	2.65	9.77	-0.11	29.6	11.38

Note. Dimensionless network density (p'), number of fractures (N), number of nonisolated fractures ($\hat{N}$), percentage of the total fractures that are in a cluster connecting inflow/outflow boundaries (%), mean of node degree distribution ($\mu(d)$), variance of degree distribution ($\sigma(d)$), Pearson assortativity coefficient ($\mathcal{P}$), graph node connectivity (n_c), and Shannon entropy. Reported values are the average of 10 network realizations.

We measure this change in connectivity using the five topological attributes presented in section 2.3.1. Interestingly, the mean of node degree distribution is roughly the same in all networks, having a value between 2 and 3. However, the variance of the degree distributions increases significantly with the higher densities. The values indicate more variability in the number of intersections in higher-density networks ($p' = 5$ and $p' = 10$). There are many fractures with few intersections and a few fractures with many intersections. Fracture degrees appear to be power law distributed, but there is an insufficient range of them to confirm such scaling; less than 2 orders of magnitude are observed. The similarity between fractures in terms of the number of intersections is measured using the Pearson assortativity coefficient ($\mathcal{P}$). In all cases, the value is less than 0, which indicates the networks exhibit disassortative mixing and that fractures of different degrees intersect. However, as the density increases, the values of $\mathcal{P}$ become closer to 0, indicating that the networks are slightly less disassortative, and more fractures of similar degree intersect. The graph node connectivity (n_c), which provides a measure of structurally imposed channelization, varies significantly with network density. All of the $p' = 2$ networks exhibit a value of 1, indicating that all of the flow must pass through a single fracture at some point in the network. As the density increases, the values increase, indicating that there are more paths between the inflow and outflow boundaries, and the flow field is less structurally constrained. Finally, the network density also influences the values of Shannon Entropy. The lowest values are observed for the smallest density values and increase monotonically. The values indicate that it is easier for flow to move from one part of the network to another in the higher density networks because these higher networks are better connected. In turn, it will be easier for flow to disperse through the network from one region to another.

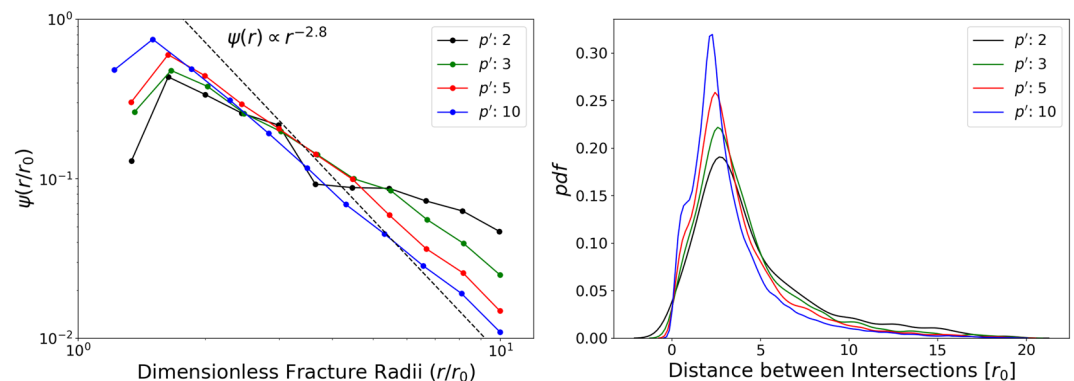


Figure 2. (Left) Distribution of fracture radii in the connected networks: $p' = 2$ —black; $p' = 3$ —green; $p' = 5$ —red; $p' = 10$ —blue. The prescribed distribution is shown as a dashed black line. Once isolated fractures and clusters are removed, there is a bias toward including more larger fractures than those prescribed while smaller fractures are removed. (Right) Distributions of distances between intersections on fracture planes. There are significantly more short distances between intersections in the higher density networks.

Table 2
Flow Channeling Metrics

p'	Geometry			Eulerian			Lagrangian	
	P_{32}	$\widehat{P}_{32}$	$P_{32}\%$	d_q	d_q/P_{32}	$d_q/\widehat{P}_{32}$	π	$\hat{\pi}$
2	0.42	0.06	13%	0.03	0.08	0.65	0.008	0.237
3	0.62	0.15	23%	0.05	0.08	0.38	0.011	0.127
5	0.96	0.63	65%	0.27	0.28	0.43	0.037	0.106
10	1.59	1.34	85%	0.80	0.50	0.60	0.056	0.105

Note. Total fracture intensity (P_{32}), connected fracture intensity ($\widehat{P}_{32}$), percent of total P_{32} in the final connected network ($P_{32}\%$), and particle-based fracture participation ratio (π). Mean values are reported for 10 realizations at each density.

The changes in density and connectivity directly impact network geometry. Figure 2 (left) shows the distribution of fracture radii in the connected networks: $p' = 2$ —black; $p' = 3$ —green; $p' = 5$ —red; $p' = 10$ —blue; prescribed distribution—dashed black line. In all networks, this prescribed distribution (1) is sampled appropriately (details not shown). However, the distribution of fractures in the connected network deviates from the prescribed one after isolated fractures and clusters are removed. Specifically, there is a bias toward including larger fractures and removing smaller ones, which is a trend observed at every density. This bias is due to the higher connectivity of larger fractures than smaller ones due to their size. Therefore, larger fractures are more likely to be retained in the connected network. Likewise, smaller fractures are poorly connected and less likely to be in the final connected cluster. As the network density increases, the differences from the prescribed distribution decrease due to increased overall connectivity in the network. These changes in density and final fracture radii distribution also influence the distance between intersections on fracture planes, the distributions of which are presented in Figure 2 (right); colors are the same as in Figure 2 (left). While the general shape of the distributions is the same, there are significantly more short distances between intersections in the higher density networks. The mean and variances in the networks are $p' = 2$: 4.56/12.24; $p' = 3$: 4.11/9.47; $p' = 5$: 3.74/8.11; $p' = 10$: 3.32/6.88. These decreasing values are the result of more intersections within the network (Table 1), and thus higher density of connections on each fracture due to higher overall network density.

3.2. Flow Channeling Measurements

Mean values of fracture intensity P_{32} , the flow channeling indicator d_q , active surface area in the generated and connected networks d_q/P_{32} , and the particle-based fracture participation ratio π are provided in Table 2. As the density increases, P_{32} in the generated network increases, which is a direct consequence of increasing the number of fractures in the network. However, $\widehat{P}_{32}$ in the connected increases in a different manner, and the percentages reflect these changes. In the $p' = 2$ networks, the connected networks include only 13% of the generated network surface area. Compare these values with the percentage of total fractures included reported in Table 1, where only 4% of the fractures are included. In combination, these values corroborate that larger fractures, which contribute more to the total surface area, make up the connected network. Compare these values with the $p' = 10$ networks where 54% of the fractures are included in the connected network, but they make up 85% of the total surface area. These discrepancies indicate that most small fractures and clusters are isolated in the high-density network, which is copacetic with the observations of the fracture radii distributions reported in Figure 2.

The changes in density and connectivity alter the percentage of surface area where there is significant flow, as measured using d_q/P_{32} and $d_q/\widehat{P}_{32}$ whose values are also reported in Table 2. As the network density increases, more of the network is included in the connected network, and thus, flow can occur therein. Thus, these values indicate that the $p' = 2$ networks have the highest degree of flow channelization within the entire network. However, the flow is not uniformly distributed throughout the connected network. The percentage of the connected network $d_q/\widehat{P}_{32}$ does not increase monotonically with increasing network density. Over 60% of the surface area has a significant amount of flow thereon in the $p' = 2$ networks. In the $p' = 3$ and $p' = 5$ networks, however, around 40% of the surface area has significant flow. The flow is fairly uniform throughout the $p' = 10$ networks.

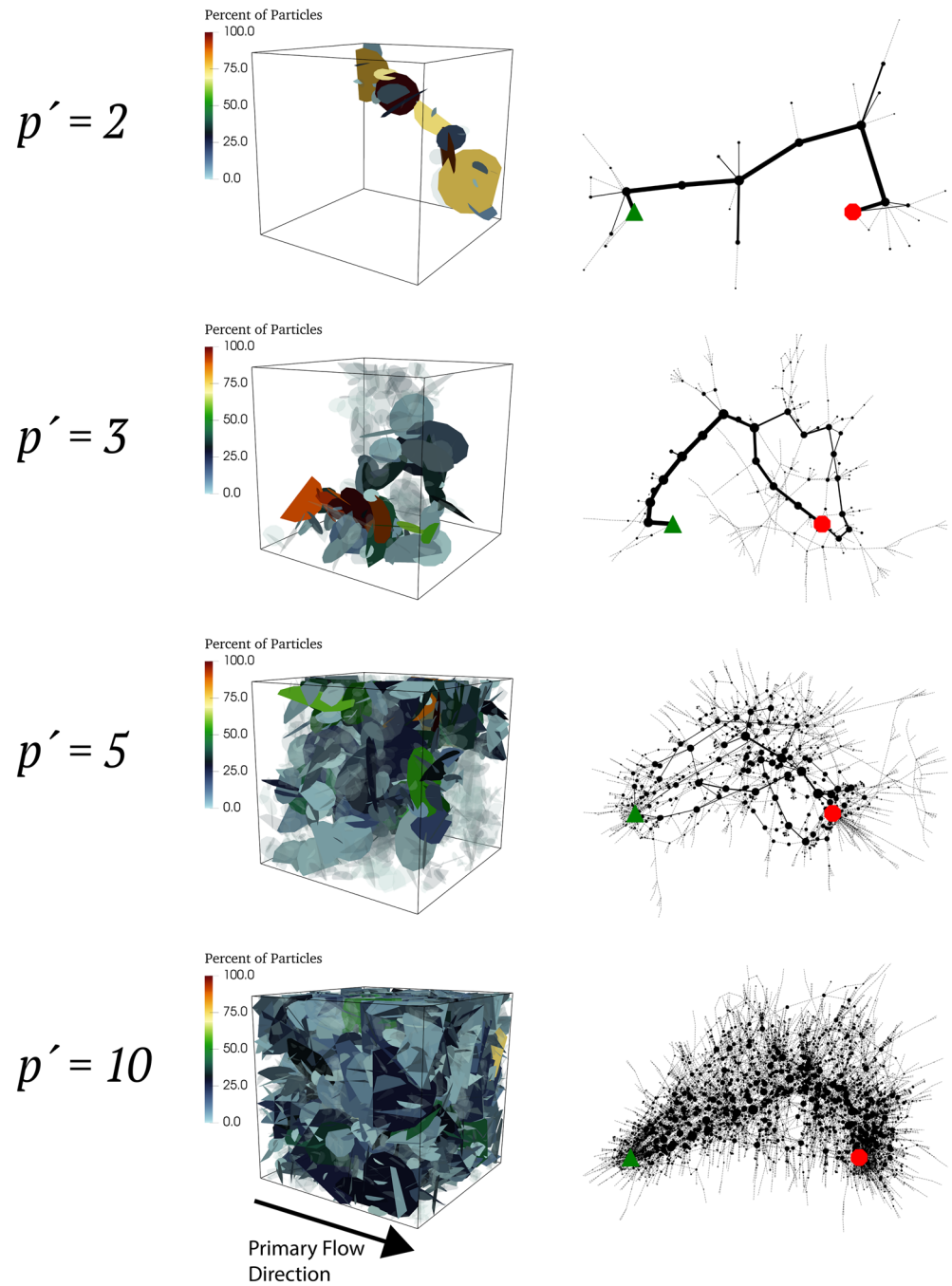


Figure 3. (Left) A single network realizations from each density. The fractures are colored by the percent of particles that pass through them. Fractures with less than 10% of the particles are semitransparent. The flow direction is from left to right. (Right) Flow topology graph representations of the networks shown on the left. The inflow boundary is indicated within a green triangle, and the outflow boundary is indicated with a red circle. The nodes in the graph represent the fractures in the network, and their size is proportional to the percent of particles that pass through them. Likewise, the edges in the graph indicate two fractures intersect, and their width is proportional to the percent of particles that move between the two fractures. Primary flow channels are seen in the lower density networks while in the higher density networks the transport of particles is fairly homogeneous.

Similar trends are observed using the particle participation ratio π . Recall that π measures the percentage of fractures by count, not surface area, that a significant number of particles traverse. Trends observed in the values computed using the generated networks are in agreement with those from the fracture counts and

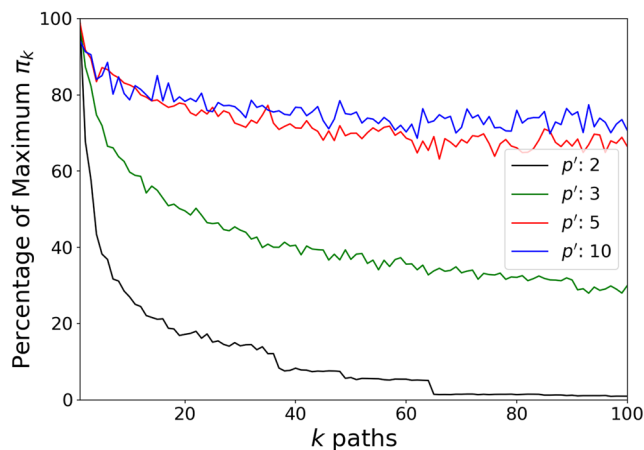


Figure 4. Integrated flow rate measure plotted as a function of the k th shortest path. Average values over each set are shown. The curve's decline rate indicates the uniformity in flow rates between paths with a rapid decline indicating more disparity between paths. Physically, the faster the decline, the more flow channeling occurring in the network.

circle. Nodes in the graph represent the fractures, and their size is proportional to the percent of particles that pass through them. Likewise, the edges in the graph indicate two fractures intersect, and their width is proportional to the percent of particles that move between the two fractures. Note that there are dead-end fractures that particles do access, as indicated by the nonzero node and edge weights. In a three-dimensional DFN, these dead-end fractures can have significant flow thereon, which is not the case in two-dimensional simulations where dead-end fractures are necessarily no-flow regions. Particles can reside on dead-end fractures for extended durations and contribute to long-term retention of solute transport that leads to delayed breakthrough times.

In the $p' = 2$ network (top row), there is one structural path from inflow to outflow boundary, and therefore, all flow and transport must pass through these fractures. This is seen in the graph representation, where a single principal channel is visible. Here, the flow is controlled by the network connectivity rather than hydrological or geometrical properties. In the $p' = 3$ network (second row from the top), there are multiple paths from the inflow to outflow boundaries, and two primary flow channels are observed. In this particular realization, there is a large portion of the network where the flow is negligible, which is in contrast to the $p' = 2$ case where there is at least some flow in most of the network. This example shows the flow is primarily constrained by the structure of the network, but variations in the local hydrological properties can also influence the structure of the flow field and which of the possible paths between the inflow and outflow carry the most flow. In the $p' = 5$ network (second row from the bottom), there are several paths between inflow and outflow boundaries, and the flow is significantly more dispersed than in the $p' = 2$ and $p' = 3$ networks. There are many fractures with a negligible amount of flow, but there are many more with some amount of flow. Moreover, it is difficult to identify primary flow paths as the differences between the flow paths are small. In the $p' = 10$ network (bottom row), there are numerous paths between the inflow and outflow boundaries, and the distribution of particles is relatively homogeneous. In this case, there are very few fractures with a negligible amount of flow, and it is not possible to identify a primary flow path.

The integrated flow rate defined in Equation 12 can be used to compare the degree of flow channeling at the different densities. The average values of π^k for the first 100 shortest paths in the four density sets are shown in Figure 4. The values have been rescaled by the value of the shortest path $k = 1$. Note that while the path lengths are monotonic, the values of π^k have slight wiggles that occur due to the inclusion of the number of nodes in the path, which is not included in the shortest path algorithm. Nonetheless, the general trend for all sets is to decline with increasing values of k . However, the rate at which that decline occurs varies significantly between the sets. In the $p' = 2$ networks, the values of π^k decline rapidly, which indicates that the number of particles in each path drops off sharply with increasing values of k . Physically, this corresponds to there being one or two principal channels and many other paths with little flow; that is, a high degree

active surface area. In the $p' = 2$ networks, there is a very small percentage of the networks that have a significant portion where transport occurs. However, this is not surprising as only a small portion of the total network is connected and accessible to particles. As density increases, these values increase monotonically. In contrast, the reported values for the connected network show the opposite trend; they decrease monotonically. Around 25% of the fractures in the $p' = 2$ networks have a significant portion of transport occurring on them. In contrast, only 10% of the fractures in $p' = 10$ networks have a significant proportion of particles passing through them.

A visual demonstration of these features is presented in Figure 3. On the left side of the figure are single network realizations from each density (only the connected network is shown). The primary direction of flow is from left to right. The fractures are colored by the percentage of particles that pass through them; fractures with less than 10% of the particles are semitransparent for clarity. The subfigures on the right side are flow topology graph representations of the networks shown on the left. The inflow boundary is indicated within a green triangle, and the outflow boundary is indicated with a red circle.

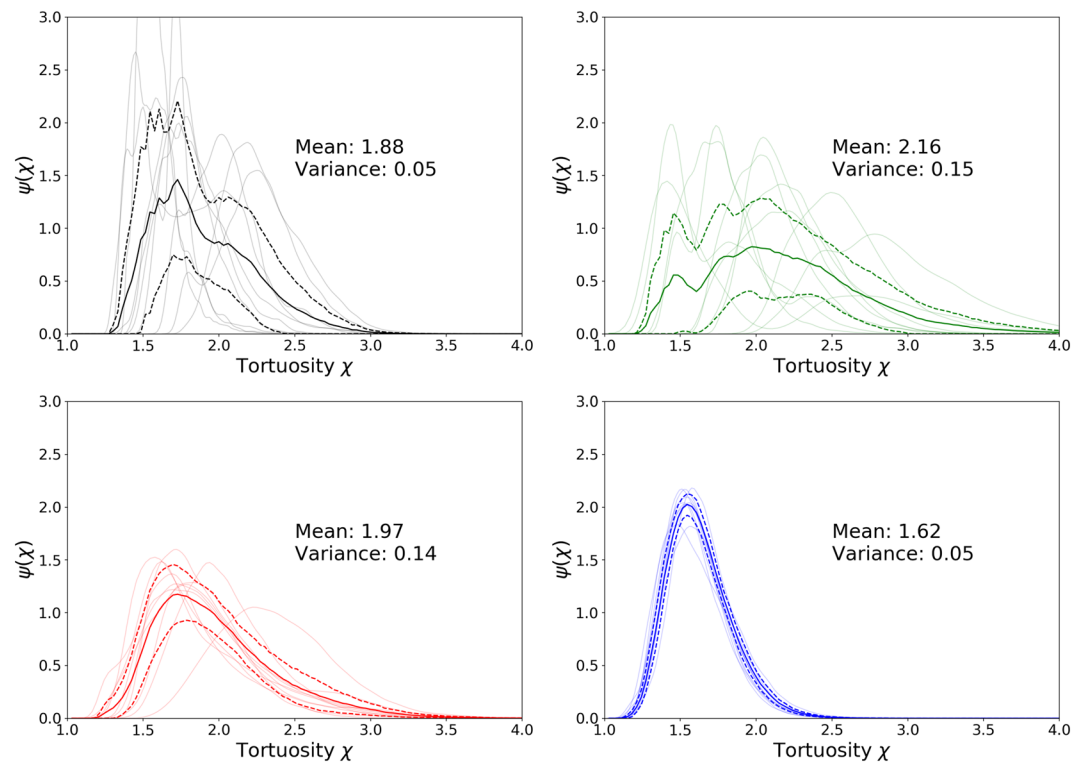


Figure 5. Tortuosity distributions from each set of networks at a given density are presented in each subfigure: $p' = 2$ —top left; $p' = 3$ —top right; $p' = 5$ —bottom left; $p' = 10$ —bottom right. Individual realizations are semitransparent, the mean value is a solid line, and the dashed lines are the 95% confidence intervals across the sets of networks. At low densities, the structure of the network constrains the pathlines due to reduced connectivity, and a unique distribution of pathlines is observed in each realization. At higher densities, the higher connectivity allows for particles not to be constrained in this way, and uniform behavior across a density set is observed.

of flow channeling occurs in these networks. In contrast, the $p' = 10$ set shows only a minor drop in π^k with increasing values of k indicating little discrepancy in the number of particles traveling on the paths, that is, a low degree of flow channeling is observed. In between these end members, we see that the degree of flow channeling, as measured by the decline rate of π^k , decreases with increasing density, which is in agreement with the visual representations presented in Figure 3.

3.3. Impact on Transport

The observations reported in the previous section indicate that structural constraints imposed by different network densities lead to varying degrees of flow channeling. In this section, we examine how these changes in flow channelization impact transport properties. We begin with the pathline tortuosities defined by Equation 10 and are shown in Figure 5: $p' = 2$ —top left; $p' = 3$ —top right; $p' = 5$ —bottom left; $p' = 10$ —bottom right. Individual realizations are semitransparent, the mean value is a solid line, and the dashed lines are the 95% confidence intervals across the sets of networks. There is a wide range of distributions observed in the $p' = 2$ networks. Some are rather tight with lower values and small variance, while others are quite broad with higher mean values. Recall that flow and transport in these networks are controlled by the network structure and all particles must pass through the same fractures. Therefore, each network structure imprints a unique distribution of pathlines. Similar behavior is observed in the $p' = 3$ networks. In this case, however, several distributions are multimodal. Here, the pathlines are primarily constrained by the network structure but can access multiple principal pathways. As the density increases, the discrepancies between the realizations within a set decreases. In the case of $p' = 5$, there are numerous paths through the network, and particles will take the paths of least resistance. The increased connectivity of the network allows particles not to be constrained by the particulars of the realization, and more uniform behavior is observed. There

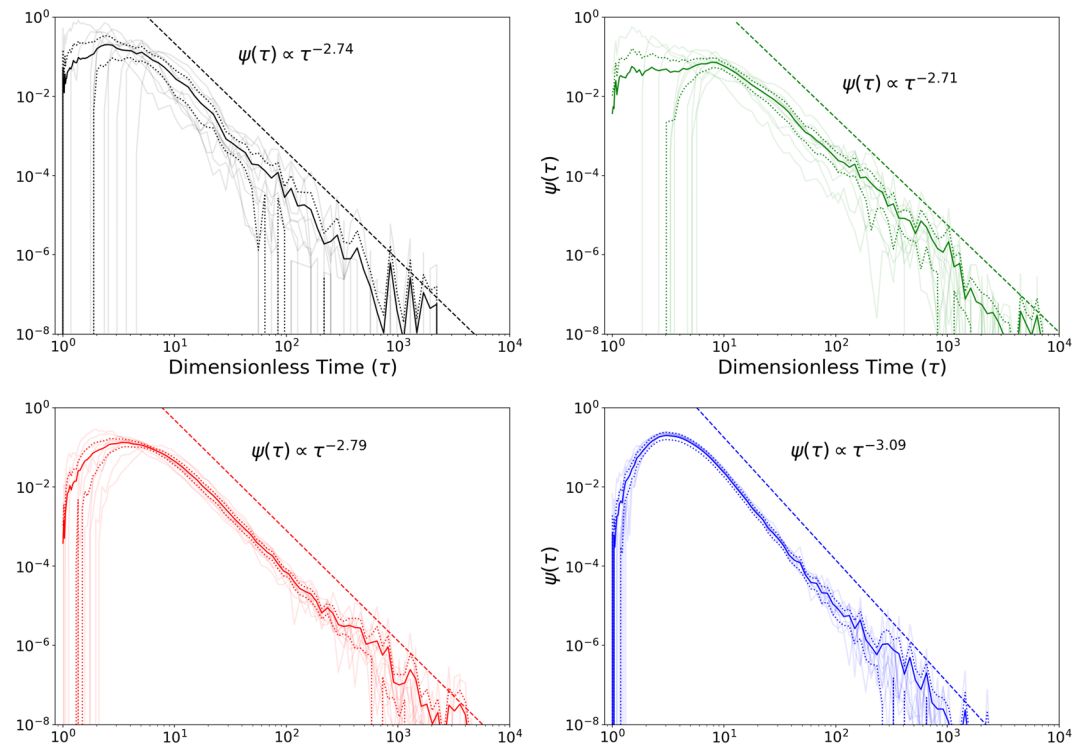


Figure 6. Particle travel time distributions from each set of networks at a given density are presented in each subfigure: $p' = 2$ - top left, $p' = 3$ - top right, $p' = 5$ - bottom left, $p' = 10$ - bottom right. Solid lines are the mean of the BTC and dotted lines are the 95% confidence intervals. Time has been non-dimensionalized by the first arrival time in the density set.

is almost no difference between realizations in the $p' = 10$ networks as the particles are not structurally constrained, and the flow field relatively homogeneous.

Finally, we consider the impact of network density on the particle travel time distributions, that is, breakthrough curves, that are presented in Figure 6: $p' = 2$ —top left; $p' = 3$ —top right; $p' = 5$ —bottom left; $p' = 10$ —bottom right. Solid lines are the mean of the breakthrough curve, and dotted lines are the 95% confidence intervals. Time has been nondimensionalized by the first arrival time in density set. The distribution of particle travel times is more concentrated to early times in the $p' = 2$ and $p' = 3$ networks than the other two network sets. These narrow peaks are the result of a high degree of flow channeling where most of the particles sample the same velocity distributions and thus have smaller differences in their travel times as they all pass through the same regions of the network. However, there are large discrepancies between individual realizations. As the density of the networks increases, the peak of the distribution broadens due to a wider range of velocities being sampled in the better-connected network and more homogeneous flow field. There is almost no difference between realizations in the $p' = 10$ networks, as the particles are not structurally constrained, and the flow field relatively homogeneous. In all density sets, the distributions exhibit power law scaling in the tail of the distribution $\psi(\tau) \propto \tau^{-\alpha}$. The $p' = 2$, $p' = 3$, and $p' = 5$ sets all show the decay rates of around $\alpha = 2.7$. However, the other highest sets show a faster decay rate of around $\alpha = 3.1$.

4. Discussion

We presented and analyzed a set of simulations designed to query how network density impacts scale flow channeling in three-dimensional fracture networks whose radii follow a power law distribution. At each density, we constructed 10 networks and characterized each of them in terms of the network topology, fracture geometry, and flow properties under steady-state conditions. We used these measurements to (i) identify flow channeling behavior in each network and (ii) link its occurrence with the structural properties of the networks. We adopted Eulerian and Lagrangian observables to quantify flow channeling. We also

demonstrated how common transport observable carry signatures of the degree of flow channeling within each network.

The principle geophysical attribute that determined the degree of flow channeling was found to be network connectivity, which depended directly upon the network density. Network densities were placed into a relative context using a dimensionless density p' based on a percolation parameter, which is used to identify the minimum number of fractures required to connect flow boundaries. Increasing the density was carried out by increasing the number of fractures in the domain while holding all other properties the same. We considered networks that were multiples of p' ranging from double to 10 times its value. Thus, the networks ranged from being very sparse to reasonably dense. As the number of fractures in the domain increased, the difference between the requested number of fractures and the size of the subnetwork that connected the inflow and outflow decreased; that is, there were fewer fractures that were inaccessible to flow due to being isolated. These observations indicated that higher-density networks were better connected, which was confirmed by the graph-based topological measurements provided in Table 1. Although the mean number of intersections on a fracture in the connected networks was similar across density values, the variance increased considerably with density. These values indicated more variability in the number of intersections in higher-density networks; that is, there were many fractures with few intersections and a few fractures with many intersections. Moreover, as the density increased, well-connected fractured intersected more well-connected fractures, as indicated by values of the Pearson assortativity coefficient. This improved connectivity resulted in networks where flow could easily disperse from one region to another, as indicated by the reported Shannon Entropy values. Such structural constraints were identified using graph node connectivity. Specifically, every one of the lowest density networks exhibited a value of 1, which indicates that all of the flow must pass through a single fracture at some point. As the density increases, these values increased, indicating that there are more paths between the inflow and outflow boundaries.

The changes in density also influenced which fractures were included in the subnetwork connecting flow boundaries and accessible to fluid flow. Specifically, we observed a bias toward including larger fractures and removing smaller ones due to the higher connectivity of larger fractures; that is, the probability of intersecting another fracture is higher for large fractures. These changes in the fracture radii distribution in the connected also influenced the distance between intersections on fracture planes; that is, there are significantly more short distances between intersections in the higher density networks, and flow can pass into and out of fractures easier. Additionally, while the fracture intensity P_{32} in the generated network increased in proportion to the density, which is a direct consequence of increasing the number of fractures in the network, it increased differently in connected subnetwork, due to the network becoming better connected and a different proportion of the generated network being included. In combination, these values all combine to show that larger fractures, which contribute more to the total surface area, make up the connected network.

As the connectivity increased with density, more of the entire network was accessible to flow. We quantified the impact of this increased connectivity on the flow field using both Eulerian and Lagrangian measurements, which are reported in Table 2. The percentage of surface area in the entire generated network where there was a significant amount increased with density, which indicated less flow channeling. This trend was also observed in terms of the cardinality of the fracture set where a significant number of particles traversed. However, the degree of flow channeling was more nuanced in the connected network. In terms of the active surface area percentage and particle-based fracture participation ratio, the highest values were observed in the lowest density networks. The flow therein was structurally constrained due to the sparsity of the network, and the flow was reasonably uniform across fractures. However, as the connected network's size increases, there was less surface area where there was a significant flow because the flow field was no longer solely determined by the network structure. On the contrary, the location of flow channels in higher-density networks was determined by a mixture of connectivity, geometry, and hydraulic properties. An example of this interplay was presented in the $p' = 3$ network shown in Figure 3 where there are multiple paths from the inflow to outflow boundaries, and a single primary flow channel splits in two. As the density increases, this combination of connectivity, geometry, and hydraulic properties trends to favor geometry and hydraulic properties because the flow is not as structurally constrained. In turn, less flow channeling is observed. Finally, the comparison of the integrated flow rate (Figure 4) defined in showed that there was some degree of flow channeling in every network.

The flow channels imprinted signatures onto Lagrangian-based observables such as the particle tortuosity and travel times. In the lowest density networks, the flow was primarily constrained to a single channel where particles pass through. Therefore, their pathline lengths (Figure 5) were all quite similar. However, there are significant differences in the distribution of tortuosities between network realizations. As the density increases, more paths are available, and the variance of the tortuosity distributions increases. In some of the $p' = 3$ networks, the distribution of tortuosity was multimodal, which corroborates the idea of multiple preferential flow paths within a DFN. However, once the density was sufficiently high, the particle trajectories are no longer structurally constrained and can travel through the geometrically/hydraulically determined paths of least resistance. This increased accessibility to regions of the network results in more uniformity across the different network realizations. Similarly, particles in the lowest densities networks showed similar travel times because they traveled through the same regions in the network (Figure 6). This channelization results in less dispersion of the particle plume and a higher/tighter peak in the travel time distribution. As the density increases, so does dispersion and lower/broader peaks in the travel time distributions are observed. Flow channeling also influences the tails of the distribution. In all density sets, power law scaling in the tail of the distribution $\psi(\tau) \propto \tau^{-\alpha}$ was observed. The differences in the decay rates reflect how particles sample the velocity field within the networks and suggest that in moderate densities, there are more discrepancies among particles.

5. Remarks

The presented observations of flow channeling and the identified link with network density/connectivity have implications for predictive modeling of flow and transport in fractured media. Foremost, the location of preferential pathways was found to be structurally determined in lower density networks, and this structure was imprinted onto transport properties, as was borne witness by the unique signatures of each sparse network realization. Thus, the use of continuum models, which do not represent such structures, for modeling transport through sparse low-permeability fractured media appears somewhat limited in applicability. Furthermore, the application of theoretical non-Fickian transport models, such as the continuous-time random walk (CTRW) or fractional-order transport models (fADE) (Benson et al., 2000; Berkowitz et al., 2006; Bolster et al., 2014; Dentz & Berkowitz, 2003; Dentz et al., 2016; Le Borgne et al., 2008; Kang et al., 2011; Neuman & Tartakovsky, 2009), to predict transport through sparse networks will also remain challenging without the inclusion of detailed information concerning the particular geophysical formation or the flow field, which are typically limited or entirely unavailable. Nonetheless, high-fidelity simulations could be used in conjunction with these models to identify which field measurements are critical to model performance. On the other hand, although the higher density networks showed more homogeneous flow fields and uniformity between realizations, they did not exhibit Fickian transport properties; for example, power law decay in the tails of travel time distributions was observed. Thus, stochastic continuum models and theoretical non-Fickian transport models could find appropriate applications in denser networks where the classical advection-dispersion equation remains inapplicable.

While we focused solely on how network density leads to the creation of preferential pathways, it must be acknowledged that flow channelization results from the interplay of the multiple length scales in fracture networks, which range from lithospheric to aperture roughness. Although this set of simulations provided some novel insight into flow channeling, the results are limited in scope by the model setup and assumptions. Thus, future studies need to test these conclusions reached here using alternative models of flow and transport in fracture networks. For example, we considered one type of DFN network, a Poisson network where fractures are stochastically generated. It has been shown that other methods to generate DFNs, for example, those that account for nucleation and growth, can result in different topologies and flow properties (Davy et al., 2010, 2013; Hope et al., 2015; Lavoine et al., 2020; Maillot et al., 2016; Thomas et al., 2020). Subsequent studies need to be performed to continue advancing our understanding of what geophysical features lead to flow channelization and seek to rank them in order of impact. The identification of how various geophysical properties leads to preferential pathway formation is a foundational step to increasing our understanding of the subsurface, and, in particular, it is a crucial step for characterizing the feedback loop of geochemical reactions within fracture networks and structural properties, which is another topic of critical importance.

Data Availability Statement

The software used for the simulations, dfnWorks, is open source and can be obtained online (at <https://github.com/lanl/dfnWorks>). Run scripts for the simulations can be found online (at <http://dx.doi.org/10.17632/dsfjdfcysn.1>).

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